ImageNet-R-50 Fresh-Parent Persistent LogT

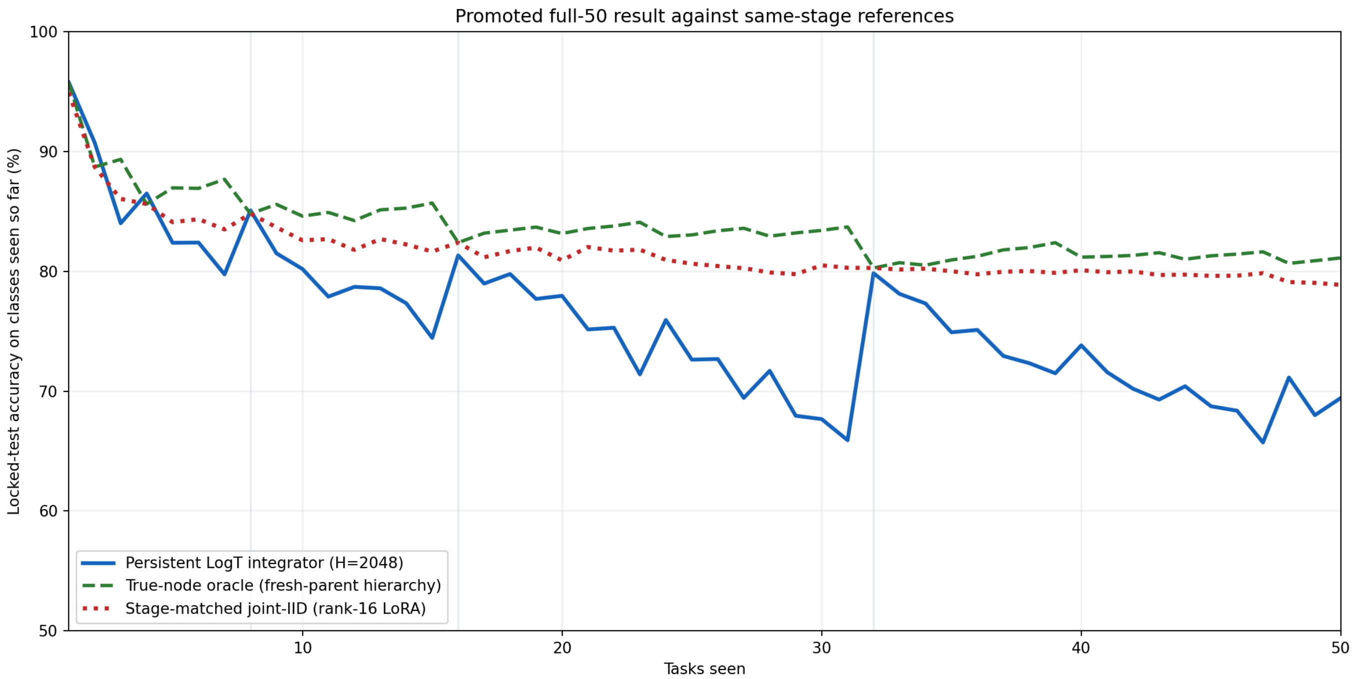
Full locked-test confirmation of the development-selected consolidation repair

Primary result. Persistent LogT reached 69.433% final / 75.709% incremental accuracy, gains of 3.867 / 2.694 points over inherited-head consolidation. The true-node oracle reached 81.117% / 83.556%. Fresh parents exactly match joint IID at every one-node power-of-two frontier. The remaining 9.433-point final and 5.922-point incremental deficits are concentrated at stages where several adapters must be combined task-free.

Why this run exists

The clean 2 x 2 x 2 factorial showed that inherited child classifier rows, not weight decay or random order, caused most of the one-node consolidation deficit. This run promotes the selected fresh-head, wd=5e-4, joint-seed recipe to every full-union parent while holding the rank-16 LoRA, topology, data, and H=2,048 persistent integrator fixed.

Primary comparison



Same locked identities and seen-class prefix at every stage. Persistent LogT is task-free; true-node routing is diagnostic and label-aware.

Tasks	Live nodes	Persistent LogT	True-node oracle	Stage-matched joint-IID
8	1	85.084%	84.803%	84.803%
15	4	74.443%	85.692%	81.649%
16	1	81.329%	82.396%	82.396%
31	5	65.907%	83.700%	80.294%
32	1	79.842%	80.276%	80.276%
50	3	69.433%	81.117%	78.867%

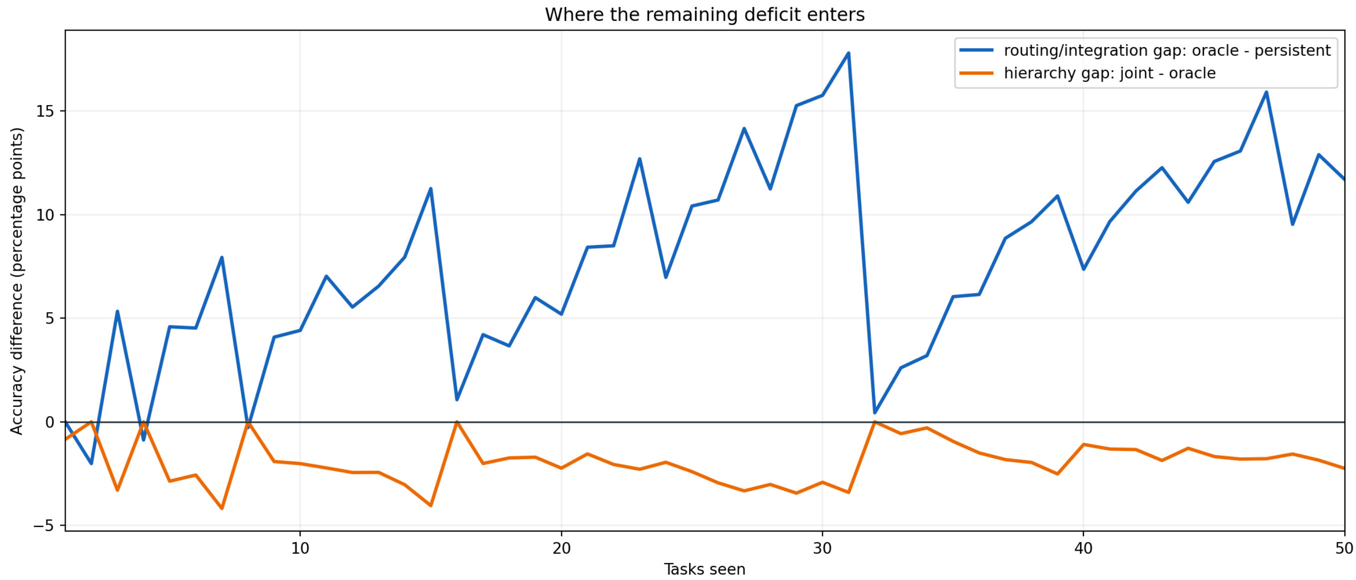
Tasks 8, 16, and 32 are one-node frontiers. Tasks 15 and 31 are the immediately preceding, maximally fragmented frontiers. Task 50 is the final three-node endpoint. These checkpoints expose the collapse-and-recovery pattern instead of sampling arbitrary stages.

Tasks	Persistent - joint	Oracle - joint
2	+2.016 pp	+0.000 pp
4	+0.877 pp	+0.000 pp
8	+0.281 pp	+0.000 pp
16	-1.067 pp	+0.000 pp
32	-0.434 pp	+0.000 pp

At tasks 2/4/8/16/32, every classifier and adapter tensor matches the corresponding fresh stage-joint model bit for bit. Some safetensors file hashes differ only in metadata. The oracle and joint accuracies are consequently identical. Persistent LogT is within 1.067 points of joint at all five except that it exceeds joint at the first three. Fresh consolidation has eliminated the parent-construction gap.

Task 31 has five independently adapted live nodes. Persistent LogT falls to 65.907%, versus 83.700% for the label-aware oracle and 80.294% for joint IID. The task-32 carry produces one joint-identical parent and persistent accuracy immediately rebounds to 79.842%. This repeated sawtooth points to adapter-dependent routing or response integration as the next main experiment.

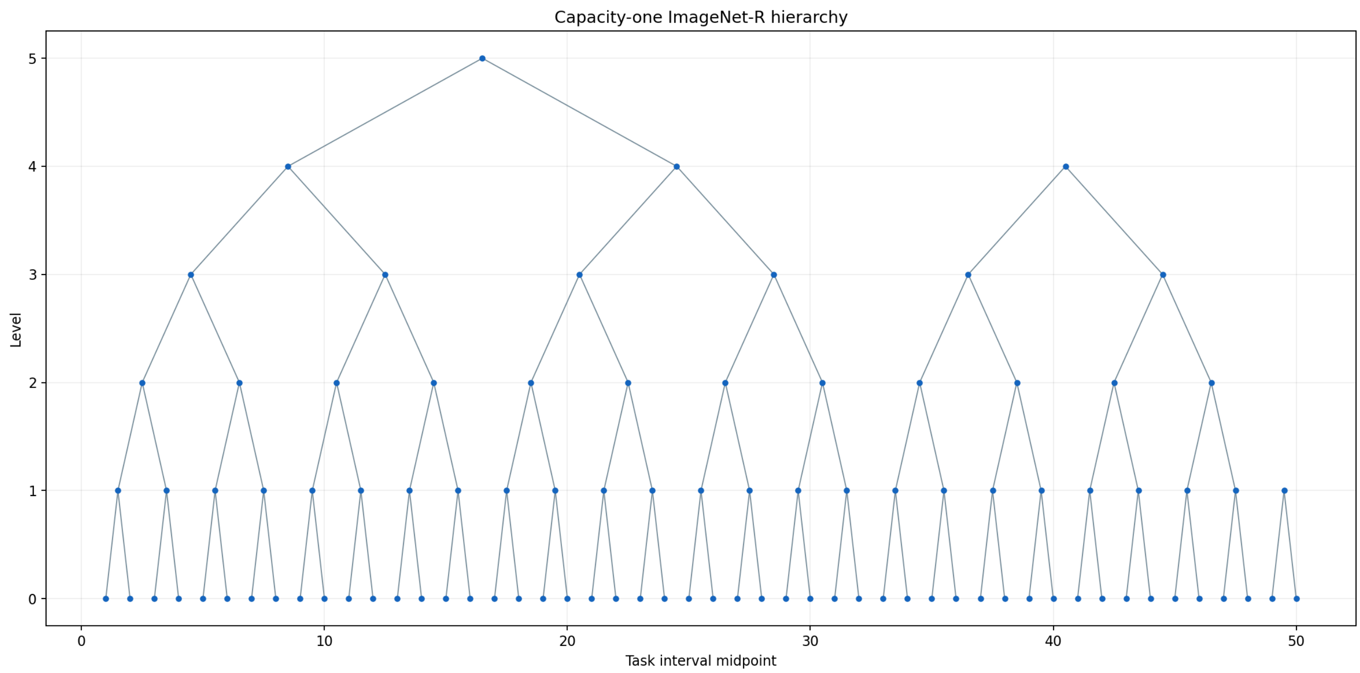
Gap decomposition



Blue: oracle minus persistent integration. Orange: stage-matched joint minus oracle; negative values mean the label-aware oracle benefits from its restricted class candidates.

Stage-matched joint IID trains a separate rank-16 QKV-plus-fc1 LoRA on only the examples available at each stage and reaches 78.867% final / 81.630% incremental accuracy. The task-50 offline joint model has the same 78.867% endpoint but a future-informed retrospective incremental score of 84.795%. Local E2-LoRA reaches 78.100% / 82.995%. The prior persistent run reached 65.567% / 73.015%. None is an execution gate.

Binary-counter lineage



Fifty arrivals create 47 carries and leave three live nodes. Power-of-two arrivals collapse the frontier to one node.

Component	Image presentations	Optimizer steps	Peak VRAM
hierarchy nodes	733,805	11,750	3.859 GiB
persistent integrator	484,988	1,024	0.083 GiB

The first workflow took about 47 minutes on the RTX 4090. An immediate resume took 3.798 seconds, did no optimizer work, and left all 170 node manifests, 170 node records, and 50 persistent checkpoints unchanged by count, nanosecond-mtime sum, and stat fingerprint.

- Head initialization: `fresh`
- Parent weight decay: `0.0005`
- Seed schedule: `joint`

The development factorial preceded this run. Hierarchy and persistent-integrator fitting completed before test behavior was opened, and the seal records zero test requests before completion. The run straddled the source commit, so node `git_commit` annotations contain either `ce066ba` or `d8924e8`; that informational field is excluded from identity, while the authoritative code manifest's material bytes exactly match `d8924e8`. This is one deterministic seed. Earlier project runs have evaluated the same locked split, although their test outcomes did not select this recipe; replication and an additional untouched protocol are needed for a publication claim.

The report directory includes CSV, JSON, and Parquet stage, power-frontier, task, and resource projections plus concise run/resume logs. A separate authenticated checkpoint-parity record preserves the bitwise tensor checks. Hash-chained behavior requests, protocol records, and the locked training seal remain in the run tree. All 68 default focused tests pass; six real-model/GPU tests stay opt-in. Large checkpoints stay local.